

Project Report On Automated Employee Onboarding & Offboarding System in ServiceNow

Project Title: Automated Employee Onboarding & Offboarding System in ServiceNow

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GitHub Repository: <https://github.com/siddhzz-siddu/ServiceNow>

Onboarding and Offboarding



Abstract

This project demonstrates the automation of employee onboarding and offboarding processes in ServiceNow using Service Catalog, Catalog Variables, Flow Designer, Approvals, and Notifications.

The system enables users (HR or employees) to raise onboarding or offboarding requests through a catalog item. It captures all required employee details and processes the request automatically using Flow Designer. The system dynamically assigns tasks to IT, Facilities, and Security teams, ensures approval from the manager, and updates the request status based on task completion.

This solution eliminates manual coordination, reduces delays, and improves visibility across departments by automating the entire employee lifecycle workflow.

Introduction

ServiceNow is a cloud-based IT Service Management (ITSM) platform that enables organizations to automate business processes efficiently. One of its most powerful features is workflow automation using Flow Designer.

Employee onboarding and offboarding are critical processes in organizations. When handled manually, they often lead to delays, miscommunication, and missed responsibilities between departments.

This project addresses these issues by:

1. Creating a Service Catalog item for onboarding/offboarding requests
2. Automating approvals using Flow Designer
3. Assigning tasks to multiple departments
4. Tracking request progress in real time
5. Sending automated notifications

Objectives

1. **Automation** – Eliminate manual handling of onboarding/offboarding
2. **Standardization** – Ensure structured workflow execution
3. **Efficiency** – Reduce delays in approvals and task handling
4. **User Experience** – Provide a dynamic and simple request form
5. **Transparency** – Enable real-time tracking of request status in real time.

Literature Review

This project is built using key ServiceNow concepts:

- Service Catalog – Interface for users to submit requests
- Catalog Variables – Capture employee details
- Catalog UI Policies – Dynamic form behavior
- Flow Designer – Automates the workflow
- Approvals – Ensures authorization by manager
- Catalog Tasks (sc_task) – Assigns work to departments
- Notifications – Sends email updates

This project integrates all these components into a unified automated system.

Methodology

The project follows a step-by-step approach:

Step 1: Service Catalog Item Creation

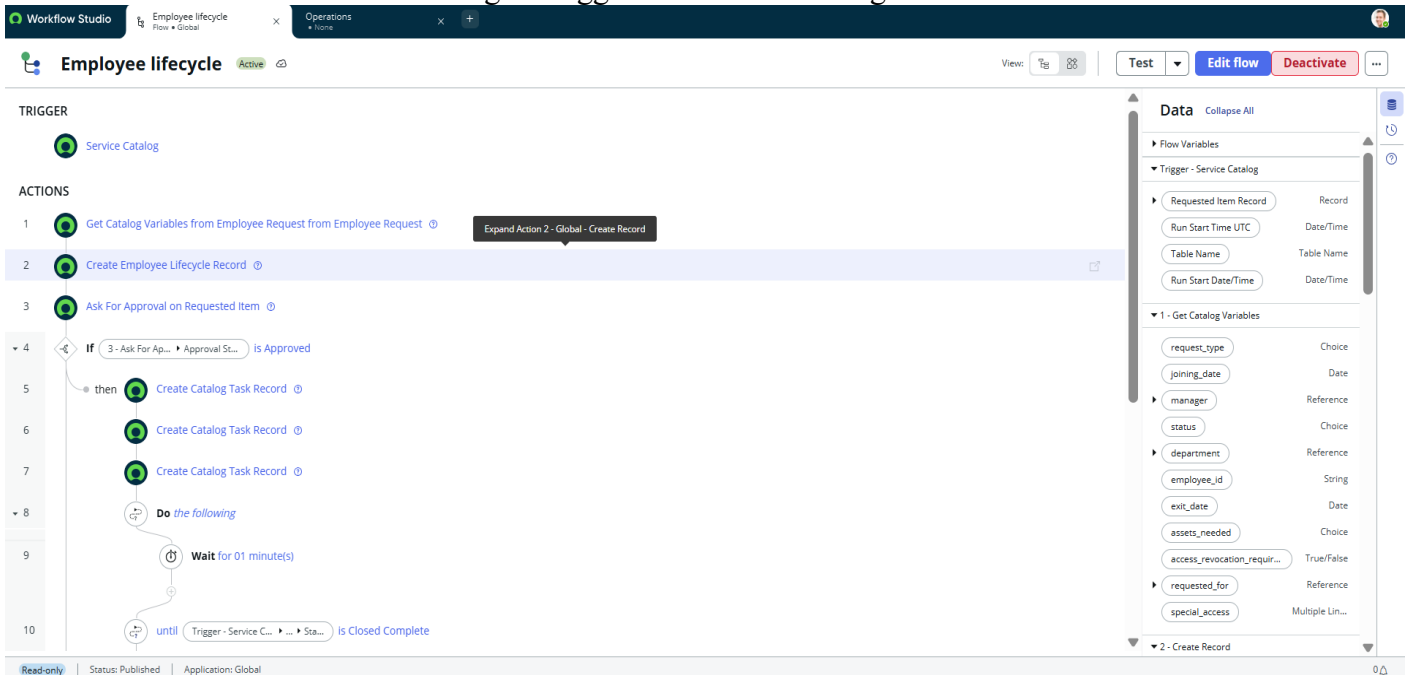
A catalog item named:

“Employee Request”

was created to capture employee onboarding/offboarding requests.

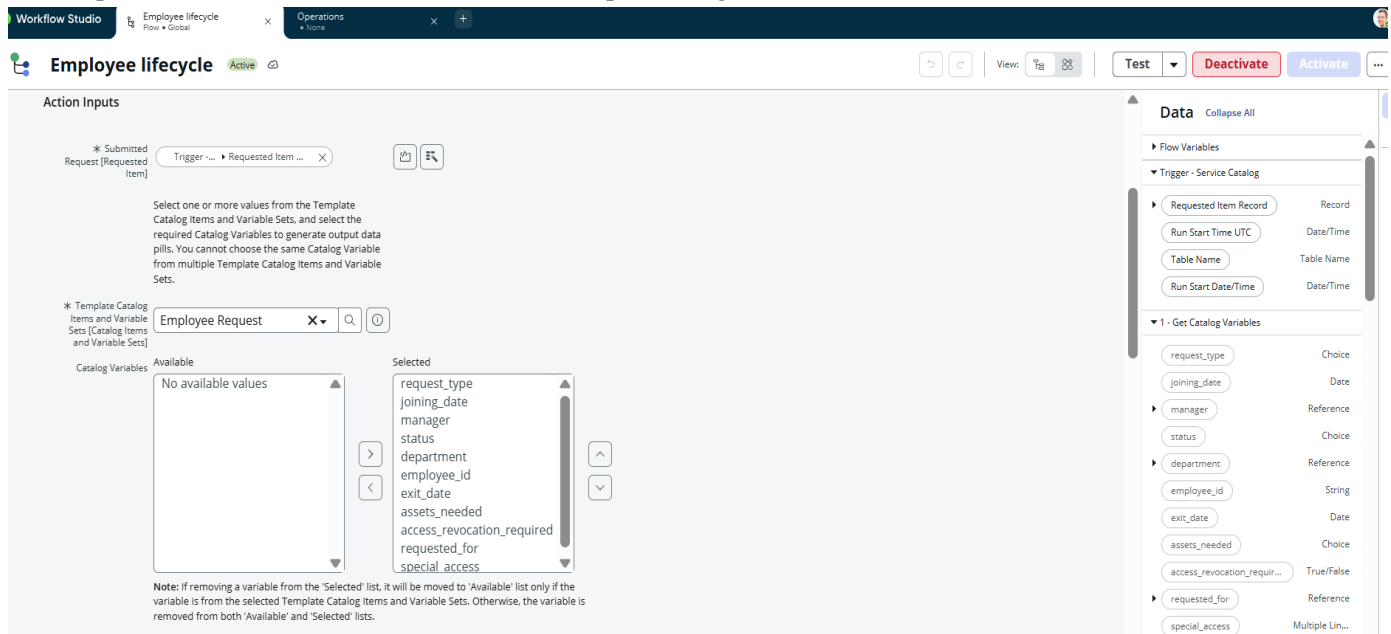
Step 3: Flow Designer – Initial Setup

A new flow was created in Flow Designer triggered when the catalog item is submitted.



Step 4: Get Catalog Variables Action

This step fetches the submitted variables for further processing.



Step 5: Create Lifecycle Record

A record is created in custom table:

Employee Lifecycle

(u_employee_lifecycle) This helps track employee request status.

Workflow Studio Employee lifecycle Flow Global Operations None

Click or hit enter to Expand ServiceNow Core Application Trigger for Service Catalog

Service Catalog

ACTIONS Select multiple

- Get Catalog Variables from Employee Request from Employee Request
- Create Employee Lifecycle Record

Action Properties

Action Create Record

Action Inputs

* Table Employee Lifecycle [u_employee...]

* Fields

Employee ID	1 - Get Catalog V... ▶ departm...
Department	1 - Get Catalog Variab... ▶ stat...
Status	1 - Get Catalog Vari... ▶ mana...
Manager	1 - Get Catalog... ▶ requested...
Requested for	1 - Get Catalog V... ▶ joining_d...
Joining date	

+ Add field value

Data Collapse All

Flow Variables

Trigger - Service Catalog

Requested Item Record

Run Start Time UTC	Date/Time
Table Name	Table Name
Run Start Date/Time	Date/Time

1 - Get Catalog Variables

request_type	Choice
joining_date	Date
manager	Reference
status	Choice
department	Reference
employee_id	String
exit_date	Date
assets_needed	Choice
access_revocation_requir...	True/False
requested_for	Reference
special_access	Multiple Lin...

Step 6: Ask for Approval

Approval is sent to:

Manager (dynamic reference from form)

The request proceeds only after approval.

Workflow Studio Employee lifecycle Flow Global Operations None

2 Create Employee Lifecycle Record

3 Ask For Approval on Requested Item

Action Properties

Action Ask For Approval

Action Inputs

* Record Trigger ... ▶ Requested Item ...

Table Requested Item [sc_req_item]

Approval Reason

Approval Field Approval

Journal Field Approval history

* Rules

Approve When:

Anyone approves 1 - Get Catalog Vari... ▶ mana...

Due Date None

Add another OR rule set

OR AND

Delete Cancel Done

Data Collapse All

Flow Variables

Trigger - Service Catalog

Requested Item Record

Run Start Time UTC	Date/Time
Table Name	Table Name
Run Start Date/Time	Date/Time

1 - Get Catalog Variables

request_type	Choice
joining_date	Date
manager	Reference
status	Choice
department	Reference
employee_id	String
exit_date	Date
assets_needed	Choice
access_revocation_requir...	True/False
requested_for	Reference
special_access	Multiple Lin...

2 - Create Record

Step 7: Conditional Flow Logic

If Approved:

1. Create Catalog Tasks:
 - IT Task → Account setup & hardware
 - Facilities Task → Workspace preparation
 - Security Task → ID & access
2. Wait until:
All tasks are Closed Complete
3. Update records:
 - Lifecycle → Completed
 - RITM → Closed Complete
4. Send completion email.

The screenshot shows the Workflow Studio interface for the 'Employee lifecycle' workflow. The 'If' condition is configured with the following settings:

- Condition Label: (empty)
- Condition 1: 3 - Ask For Ap... → Approval St... is Approved
- Buttons: Add another condition set(OR), Delete, Cancel, Done

The right sidebar shows the 'Data' panel with the following variables:

- Flow Variables
- Trigger - Service Catalog
 - Requested Item Record (Record)
 - Run Start Time UTC (Date/Time)
 - Table Name (Table Name)
 - Run Start Date/Time (Date/Time)
- 1 - Get Catalog Variables
 - request_type (Choice)
 - joining_date (Date)
 - manager (Reference)
 - status (Choice)
 - department (Reference)
 - employee_id (String)
 - exit_date (Date)
 - assets_needed (Choice)
 - access_revocation_requir... (True/False)
 - requested_for (Reference)
 - special_access (Multiple Lin...)
- 2 - Create Record

If Rejected:

- Update Lifecycle → Rejected
- Send rejection email to requester

The screenshot shows the Workflow Studio interface for the 'Employee lifecycle' workflow. The 'Else' branch is configured with the following steps:

13. Else
14. Update Requested Item Record
15. Update Requested Item Record
16. Send Email

The right sidebar shows the 'Data' panel with the following variables:

- Flow Variables
- Trigger - Service Catalog
 - Requested Item Record (Record)
 - Run Start Time UTC (Date/Time)
 - Table Name (Table Name)
 - Run Start Date/Time (Date/Time)
- 1 - Get Catalog Variables
 - request_type (Choice)
 - joining_date (Date)
 - manager (Reference)
 - status (Choice)
 - department (Reference)
 - employee_id (String)
 - exit_date (Date)
 - assets_needed (Choice)
 - access_revocation_requir... (True/False)
 - requested_for (Reference)
 - special_access (Multiple Lin...)
- 2 - Create Record

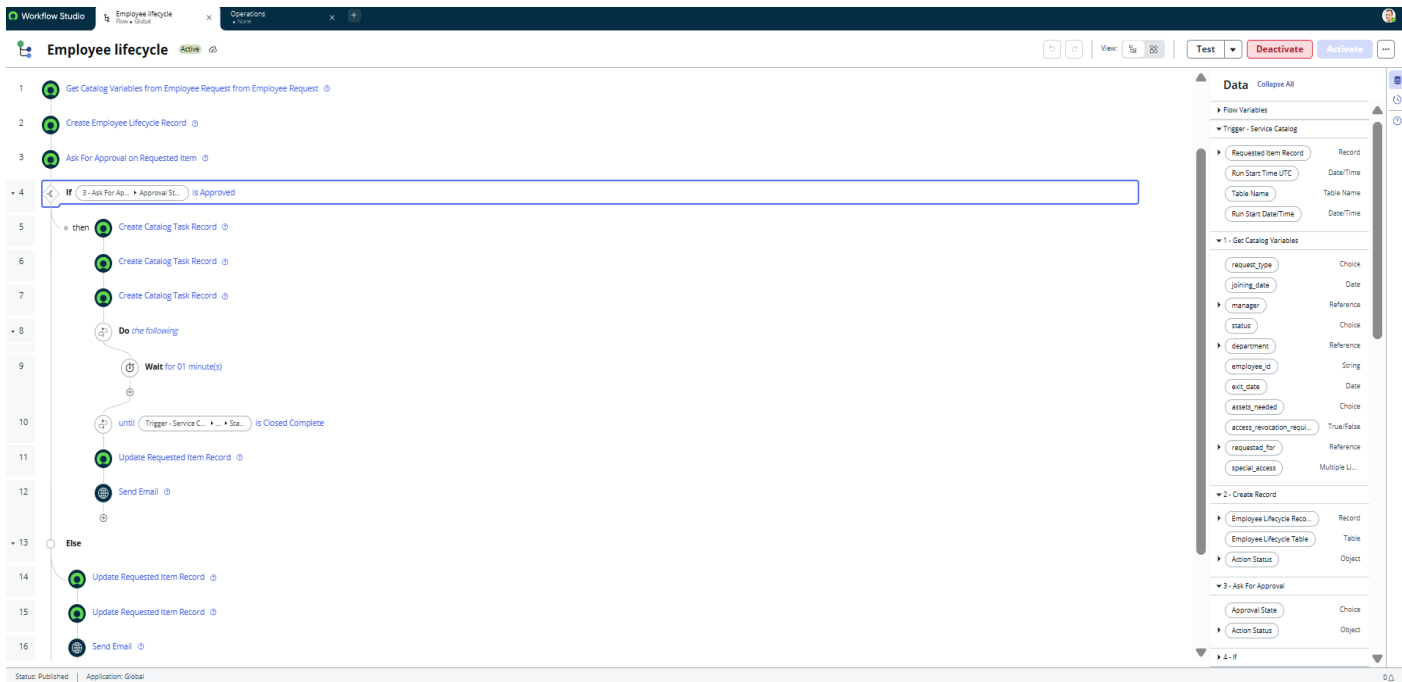
Step 8: Wait Condition Logic

The flow uses a loop mechanism to wait until:

All sc_task records reach Closed Complete

This ensures proper coordination between departments.

Step 9: Final Flow Design



The complete workflow is now automated end-to-end.

Results

1. Users can submit onboarding/offboarding requests easily
2. Requests are automatically processed and tracked
3. Tasks are assigned to relevant departments
4. Approvals and notifications are automated
5. Manual effort is significantly reduce

Conclusion

This project successfully demonstrates the automation of employee onboarding and offboarding using

ServiceNow. By integrating Service Catalog, Flow Designer, Approvals, and Notifications, the system ensures:

- Faster processing
- Improved coordination
- Better visibility
- Reduced manual errors

This solution can be extended to other HR and IT service processes.

Future Enhancements

1. Integration with Active Directory for account creation
2. SLA tracking for task completion
3. Virtual Agent (chatbot) integration
4. Dashboard for HR analytics and reporting

References

1. ServiceNow Documentation – <https://docs.servicenow.com>
2. GitHub Project Repository – <https://github.com/siddhzz-siddu/ServiceNow>